

Software Reverse Engineering Course

- Site: OnHexGroup.ir
- Youtube: @onhexgroup
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Windows
(x86 / x64)

By: OnhexGroup

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CPU

- Processing Unit in a Computer
- The CPU executes machine code
- Different processors have different architectures and machine-code formats
- Architectures differ in execution, optimizations, resource usage, etc.

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Intel x86 Architecture

- Related to the Intel 8086 processor family
- The first 16-bit model evolved from the 8-bit 8080
- Expanded into multiple processor models:
 - 8086 – 80186 – 80286 – ...

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x64 Architecture

- Intel developed a new architecture called Intel64 / IA-64 / Itanium
- AMD focused on extending x86 and developed AMD64 (x86-64)

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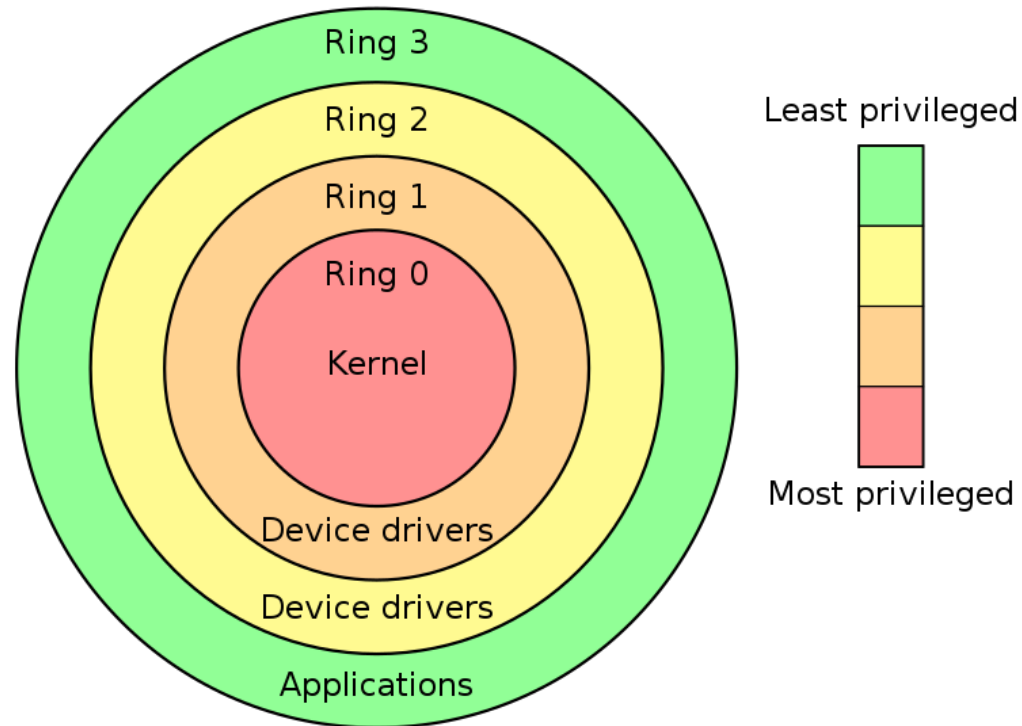
CPU Modes in Intel Processors

- The x86 family has two operating modes:
 - **Real Mode**
 - The first operational mode of the x86 processors
 - Executes 16-bit instructions
 - Still exists for compatibility (e.g., reboot/BIOS)
 - Programs have unrestricted access to the address space
 - **Protected Mode**
 - Introduced after the 80286
 - Supports virtual memory, paging, access control, and more
 - Modern operating systems run in this mode

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Access Control

- Modern operating systems use a concept called Rings for privilege control:



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Course Focus

- In this course we will cover:
 - Intel x86 family
 - Both 32-bit and 64-bit versions
 - Operating in Protected Mode

References

- https://en.wikipedia.org/wiki/Machine_code
- https://en.wikipedia.org/wiki/Computer_architecture
- <https://en.wikipedia.org/wiki/X86>
- <https://en.wikipedia.org/wiki/X86-64>
- https://en.wikipedia.org/wiki/Protected_mode
- https://en.wikipedia.org/wiki/Real_mode